Servo Arm Control System

1. Product Overview

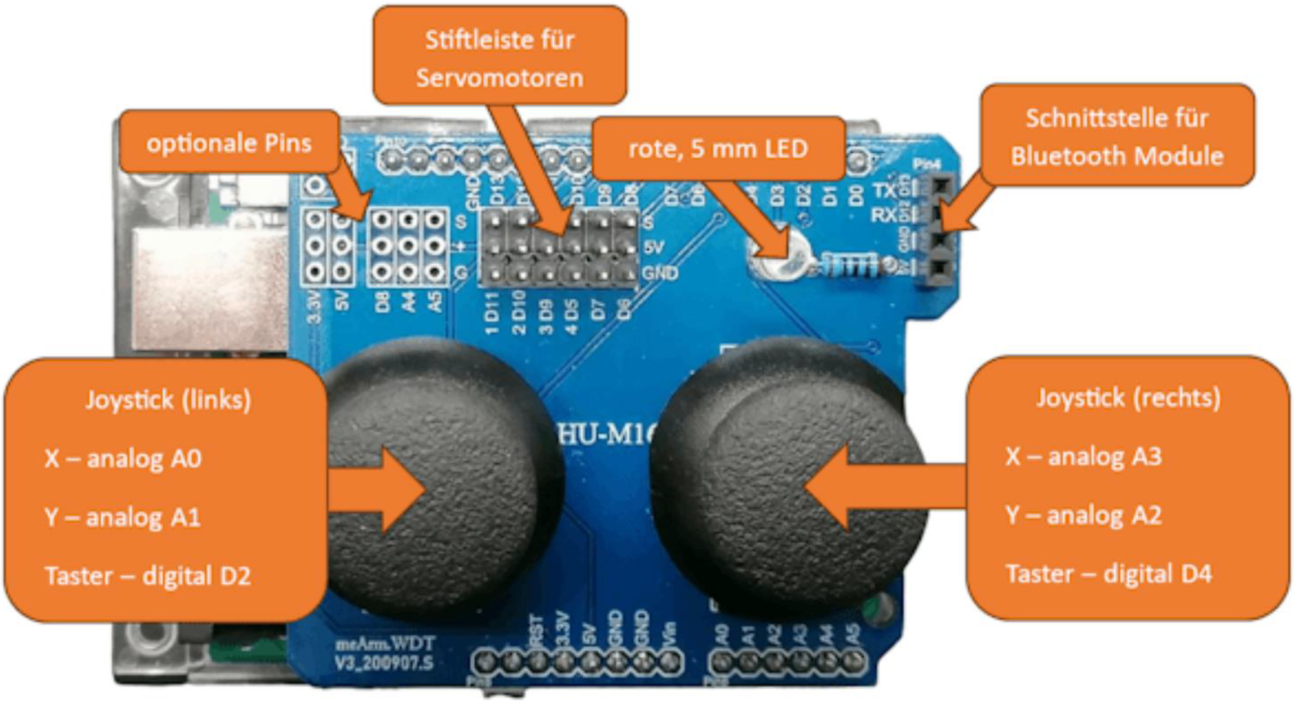
A complete 5-servo robotic arm control system with three control methods:

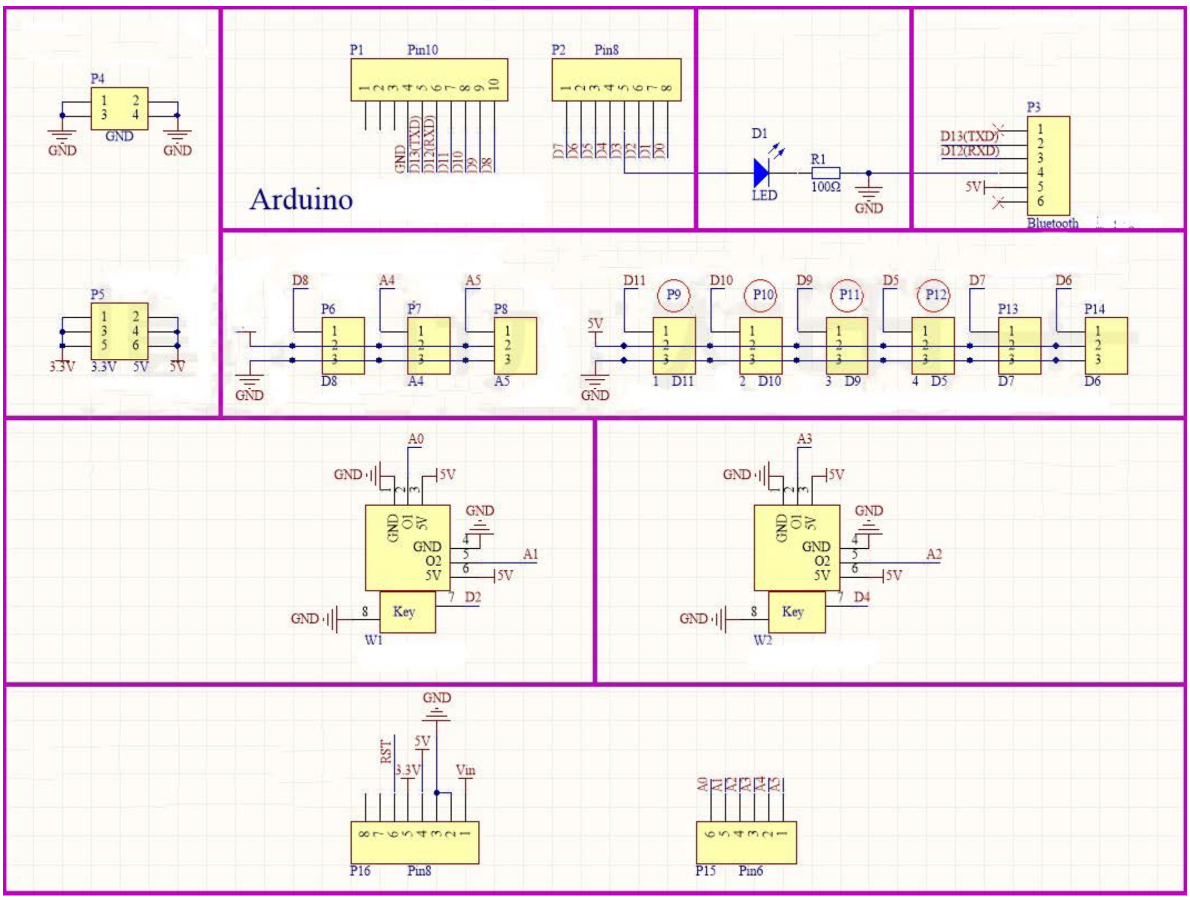
- Joystick Control - Dual joysticks for real-time control
- Serial Commands - Arduino IDE Serial Monitor
- Python GUI - Computer interface with visual controls

2. Hardware

- Arduino UNO R3 (main controller)
- 2 dual-axis joystick modules
- 5 MG995 servo motors
- Mechanical arm components

3. Pin Definition





4. Control Logic

Control Mapping Table:

Control	Pin	Servo	Range	Function
Joystick 1 X	A0	Servo 0	0-180 deg	X-axis
Joystick 1 Y	A1	Servo 1	0-180 deg	Y-axis
Joystick 1 Btn	D2	Servo 4	0-90 deg	+2 deg/press
Joystick 2 X	A3	Servo 2	0-180 deg	X-axis
Joystick 2 Y	A2	Servo 3	0-180 deg	Y-axis
Joystick 2 Btn	D4	Servo 4	0-90 deg	-2 deg/press

5. Arduino Setup

1. Open servo_control.ino in Arduino IDE
2. Select Board: Arduino UNO
3. Select COM port
4. Upload sketch
5. Open Serial Monitor (9600 baud)

6. Serial Commands

0 90 - Set Servo 0 to 90 deg
1 45 - Set Servo 1 to 45 deg
2 90 - Set Servo 2 to 90 deg
3 90 - Set Servo 3 to 90 deg
4 45 - Set Servo 4 to 45 deg (max 90)
90 - Set ALL to 90 deg
s - Show status
h - Show help

7. Python G

Requirements: Python 3.x, pyserial

Setup: pip install pyserial

Run: python servo_control.py

8. Safety Warning

Power Supply:

After uploading code, switch from computer USB to external power supply (9V 2A or higher).

MG995 servos require adequate power.

Overload Protection:

Do not apply excessive force. Servo stall causes current surge and board restart.

Support:

GitHub: https://github.com/490437477-sys/servo-arm-control